

Data Dictionary for Extended Neuropath, v1 (July 2024 release)

Variables were selected and coded to agree with the ADSP Phenotype Harmonization Consortium Neuropath Data Dictionary (<https://www.vmacdata.org/adsp-phc/data/data-dictionary>); some additional variables were added by study neurologists (alt_LEWY_full, CDLB, alt_HS_L, Lacunes)

For additional details regarding derivation of specific variables, see the AMP-AD Neuropathology Data Extraction document compiled by Mayo Clinic (20231121_AMP-AD_U01_EXPOSOME_Neuropathology_Data_Extraction_GY.pdf)

individualID	identifying string linked to a particular individual (example: R1234567)
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dataContributionGroup	The research group who contributed the sample
	Columbia
	Emory
	Mayo
	MSSM
	Rush

cohort	The initial study group population to which the individual belonged
	AA
	Banner
	Biggs Institute Brain Bank
	CLINCOR
	Emory
	LATC
	MAP

	MARS
	Mayo Clinic
	Mt Sinai Brain Bank
	NY Brain Bank
	UPenn
	ROS
	UFL

LEWY_full	PD Braak Staging (1,2)
	0 = None
	1 = Brainstem predominant
	2 = Limbic/amygdala (transitional)
	3 = Neocortical (diffuse)
	4 = Olfactory bulb or Region unspecified
	NA = missing or unknown

alt_LEWY_full	PD Braak Staging, alternate version (Modified classification of Kosaka criteria (1,2), determined by Dr. Dennis Dickson at Mayo Clinic)
	0 = None
	1 = Olfactory bulb only
	2 = Amygdala only
	3 = Brainstem predominant
	4 = Limbic (transitional)
	5 = Neocortical (diffuse)
	NA = Missing or unknown

LEWY_grp	PD Braak (grouped)
	0 = None
	1 = Brainstem predominant
	2 = Transitional, diffuse, Amygdala only, olfactory bulb predominant, or region unspecified
	NA = Missing or unknown

LEWY_ANY	PD Braak (dichotomous)
	0 = None
	1 = Brainstem, transitional, diffuse, Amygdala only, olfactory bulb predominant, or region unspecified
	NA = missing or unknown

CDLB	Likelihood of Lewy Body clinical syndrom reported based on the Third report of the DLB consortium (3)
	1 = Low/No clinical significance
	2 = Intermediate
	3 = High
	NA = missing or unknown

INFA	Infarcts and lacunes. Reports of anoxic ischemic injury will not be considered within the infarcts and lacunes lesions.
	0 = No
	1 = Yes
	NA = missing or unknown

HEMOR	Hemorrhages
	0 = No
	1 = Yes
	NA = missing or unknown

MICR	Microinfarcts
	0 = No
	1 = Yes
	NA = missing or unknown

WMRCAL	White matter rarefaction (cerebral arteriosclerotic leukoencephalopathy). Simplified version to combine WMR (v10 of NACC) and ART (LOAD, Spanich cohort, NACC v9 and later, and more)
	0 = No
	1 = Yes
	NA = missing or unknown

WMR	White matter rarefaction
	0 = Absent
	1 = Mild
	2 = Moderate
	3 = Severe
	NA = missing or unknown

WBVD	Whole brain vascular disease
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	0 = No
	1 = Yes
	NA = missing or unknown

HS_L	Hippocampal sclerosis (with laterality)
	0 = None
	1 = Unilateral
	2 = Bilateral
	3 = Present, Location unknown
	NA = missing or unknown

HS_S	Hippocampal sclerosis (simplified)
	0 = No
	1 = Yes
	NA = missing or unknown

alt_HS_S	Unilateral Hippocampal Sclerosis. Infarcted and minimal or focal neuronal loss in the Hippocampus is most often associated with TDP-43 so it will be not considered as presence of Hippocampal sclerosis. Neuronal loss epilepsy-related (endplate) or Lewy-related (CA2) will be not considered as presence of Hippocampal sclerosis. Definition by Dr. Dennis Dickson at Mayo Clinic
	0 = Absence: Normal neuronal population or Focal/Patchy neuronal loss in all sectors of Ammon's horn or Sommer's sector.
	1 = Presence: Marked/Severe/Extensive/Moderate neuronal loss in all sectors of Ammon's horn or Sommer's sector.

	NA = missing or unknown
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TDP43	TDP Whole Brain (Presence). Calculated using the categories in TDP_3; if any of the amygdala, hippocampus, entorhinal, or neocortex are positive, it is “yes.” If assessed and negative for the above, it is “no.” <table><tr><td>0 = No</td></tr><tr><td>1 = Yes</td></tr><tr><td>NA = missing or unknown</td></tr></table>	0 = No	1 = Yes	NA = missing or unknown
0 = No				
1 = Yes				
NA = missing or unknown				

TDP_3	TDP-3 Component
	0 = None
	1 = Amygdala
	2 = Hippocampus or Entorhinal
	3 = Neocortex
	NA = missing or unknown

TDP_5	TDP Stage
	0 = None
	1 = Brainstem/Spinal Cord
	2 = Amygdala
	3 = Hippocampus
	4 = Entorhinal
	5 = Neocortex
	NA = missing or unknown

CVD_S	Cerebrovascular Disease (simplified). Composite variable indicating whether at least one of the contributing indicators (CVD_ATH_ANY, CVD_ART_ANY, CVD_CAA_ANY) are 1; else if at least one contributing indicator was evaluated as 0, CVD_S is 0; only missing if all indicators are missing <table><tr><td>0 = No</td></tr><tr><td>1 = Yes</td></tr><tr><td>NA = missing or unknown</td></tr></table>	0 = No	1 = Yes	NA = missing or unknown
0 = No				
1 = Yes				
NA = missing or unknown				

CVD_ATH	Atherosclerosis
	0 = None
	1 = Mild
	2 = Moderate
	3 = Severe
	NA = missing or unknown

CVD_ATH_ANY	Atherosclerosis (indicator)
	0 = None
	1 = Mild, Moderate, or Severe
	NA = missing or unknown

CVD_ART	Arteriolosclerosis
	0 = None
	1 = Mild
	2 = Moderate

	3 = Severe
	NA = missing or unknown

CVD_ART_ANY	Arteriolosclerosis (indicator)
	0 = None
	1 = Mild, Moderate, or Severe
	NA = missing or unknown

CVD_CAA	Cerebral Amyloid Angiopathy (CAA)
	0 = None
	1 = Mild
	2 = Moderate
	3 = Severe
	NA = missing or unknown

CVD_CAA_ANY	Cerebral Amyloid Angiopathy (CAA) (indicator). <i>Mild Cerebral amyloid angiopathy reported as (0 to 1+) in the Mayo Clinic Brain banks refers to Cerebral amyloid angiopathy that affects leptomeninges; in that case it will be considered as presence.</i>
	0 = None
	1 = Mild, Moderate, or Severe
	NA = missing or unknown

BRAINWT	Brain weight in grams (numeric)
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Lacunes_MSSM	Presence or absence of lacunes (MSSM only)
	0 = No
	1 = Yes
	NA = missing or unknown

References

- 1) Kosaka, K., et al., Diffuse type of Lewy body disease: progressive dementia with abundant cortical Lewy bodies and senile changes of varying degree--a new disease? Clin Neuropathol, 1984. 3(5): p. 185-92.
- 2) McKeith, I.G., et al., Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. Neurology, 2017. 89(1): p. 88-100.
- 3) McKeith, I.G., et al., Diagnosis and management of dementia with Lewy bodies. Third report of the DLB consortium, 2005. 65(12): p. 1863-1872.